

VTU Belagavi

REGULAR EXAMINATION ? 2025

Cryptography and Network Security ? Regular Examination, 2025

Subject Code: CS603 Semester: 6 Regulation: R18

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Classical Encryption)

1. Write short notes on Classical Encryption. [12 Marks]
2. Define Classical Encryption. Explain its significance in Cryptography and Network Security with a suitable example. [12 Marks]
3. List the advantages and limitations of Classical Encryption. Explain any two in detail. [12 Marks]

SECTION B (Unit 2: Public Key Cryptography)

4. List the advantages and limitations of Public Key Cryptography. Explain any two in detail. [12 Marks]
5. Discuss the engineering applications of Public Key Cryptography. [12 Marks]
6. Derive the expression related to Public Key Cryptography and explain each term. [8 Marks]

SECTION C (Unit 3: Hash Functions and Message Authentication)

7. Compare and contrast the key concepts involved in Hash Functions and Message Authentication. [12 Marks]
8. Write short notes on Hash Functions and Message Authentication. [8 Marks]
9. Discuss the engineering applications of Hash Functions and Message Authentication. [12 Marks]

SECTION D (Unit 4: Network Security Protocols)

10. Explain Network Security Protocols with a neat diagram/sketch and a suitable example. [12 Marks]
11. How is Network Security Protocols implemented? Give an algorithm / procedure and explain. [12 Marks]
12. Compare and contrast the key concepts involved in Network Security Protocols. [12 Marks]

SECTION E (Unit 5: Digital Signatures and Certificates)

13. How is Digital Signatures and Certificates implemented? Give an algorithm / procedure and explain. [12 Marks]
14. Solve a problem related to Digital Signatures and Certificates. Show all the steps

clearly. [12 Marks]

15. Compare and contrast the key concepts involved in Digital Signatures and Certificates. [12 Marks]

*** End of Question Paper ***